Part D: Reflection Questions

i. Which wireless/WAN technology choice in your design was hardest to justify, and why?

The Windhoek–South Africa link was the hardest to pin down. Unlike the Windhoek–Walvis Bay corridor, where I could point to real, well-documented fibre/MPLS infrastructure along the B2 route, or the global link, where satellite is clearly the only realistic option, the cross-border link sits in a grey area — it could plausibly run over leased fibre, MPLS via a regional provider, or a VPN over the internet, and each choice has different cost and reliability trade-offs that aren't obvious without pricing data I didn't have access to. I ended up justifying it based on submarine cable routing (WACS/Equiano) rather than a specific commercial product, which felt more defensible than guessing at a provider. - **Allen Lincoln Pietersen**

ii. How does Namibia's geography and population density shape the WAN/wireless choices you made for the Windhoek–branch link, compared to a similar link in a densely populated country?

Namibia's low population density is really the reason the Windhoek–Walvis Bay link could realistically use terrestrial fibre/MPLS at all — that corridor happens to follow a well-serviced highway route between two of the country's larger towns. If Kalahari Logistics needed a branch somewhere more remote, satellite would likely be the only viable option, the same way Telecom Namibia's Satlink VSAT service exists specifically because laying fibre to sparsely populated areas isn't commercially viable. In a densely populated country, that same link would almost certainly just be fibre without a second thought, because the cost per connected customer is so much lower. - **Allen Lincoln Pietersen**

iii. What cross-border challenges did you identify for the SADC link, and how might regional cooperation reduce these?

The clearest challenge is regulatory fragmentation — CRAN and ICASA operate independently, so a WAN provider serving both Namibia and South Africa typically needs separate contracts and SLAs in each country rather than one seamless regional service. Roaming costs are a related issue, though this is actually improving: Namibia and Botswana's SADC One Network Area agreement, which has since expanded to several other member states, shows that harmonised roaming is achievable when regulators coordinate. If South Africa and the rest of SADC eventually move toward the same kind of flat-roaming model the EU uses, it would meaningfully lower the operational friction for a company running a WAN across the region. - **Emensia Kahiriko**

iv. How did the global case study you researched change your understanding of what is technically possible versus what is currently deployed in Namibia?

Researching the SD-WAN case studies made it clear that mature markets have already moved past MPLS as the default enterprise WAN technology, using it mainly as a fallback rather than the primary link. Namibia and South Africa are still largely in an MPLS-first stage, and Namibia's own 5G and LEO satellite rollouts are quite recent by comparison. It reframed my thinking — the gap between Namibia and global practice isn't about the technology existing yet, it's about deployment timing and market maturity, which is a different and more encouraging problem than a lack of access to the technology itself. - **Emensia Kahiriko**

v. What did using GitHub as a team teach you about version control and collaborative workflows in a real IT project?

Working with GitHub as a group taught us why version control matters in real projects. Splitting the repo into clear folders made it obvious who was responsible for what, and committing regularly meant we always had a working version to fall back on if something went wrong. We also ran into a merge conflict early on, which taught us to communicate before working on the same file at the same time. Overall, it showed us that version control isn't just about backing up work — it's about making a team's contributions visible and easy to track. - **Tuzeraje C G Ndjavera**

vi. If Kalahari Logistics doubled its number of branch offices next year, what would you change about your design to scale it?

The current design uses static routes, which works fine for four sites but wouldn't scale well — doubling the branch count would mean manually adding routes on every router for every new site, which gets error-prone fast. I'd move to a dynamic routing protocol like OSPF so new sites could be added with minimal reconfiguration elsewhere in the network. I'd also reconsider the addressing scheme, since the current /24 per site and /30 per link scheme was sized for four sites and would need a clearer allocation plan to avoid running out of address space or creating overlapping subnets as more branches get added. - **Tuzeraje C G Ndjavera**

vii. What was the most valuable lesson your group learned from this assignment, and how does it connect to a real-world career in networking?

The most useful part was seeing how much a WAN design decision depends on context that has nothing to do with the technology itself — population density, regulatory regimes, submarine cable geography, even ownership rules for licensing. It's a reminder that real network design work isn't just picking the "best" technology in the abstract, it's picking the technology that's actually viable given the constraints of a specific place and business, which is exactly the kind of judgment call a working network engineer or consultant has to make. - **Allen Lincoln Pietersen**